

Beyond LLMs: The Rise of Tabular Foundation Models

A deep dive into TabPFN-2.5 — the transformer that wants to replace
XGBoost

Arshan Bhanage • Deep Learning Short Story • 2025

Motivation

Foundation models have revolutionized text (GPT), images (DALL·E), and code (Copilot)

But the most common data type in the real world is tabular data — spreadsheets, CSVs, database rows

XGBoost & gradient-boosted trees have dominated tabular ML for a decade

Can foundation models finally challenge this dominance?

TabPFN-2.5 claims: YES — matching 4-hour AutoGluon ensembles in a single forward pass

Why Tabular Data Matters

Healthcare

Patient records, lab results, diagnoses

Finance

Transactions, credit scores, risk models

Manufacturing

Sensor data, quality metrics, supply chain

Retail

Customer data, purchases, inventory

Tabular data powers virtually every business decision on the planet — any improvement has outsized impact

What Are Foundation Models?

A single model pre-trained on massive data that adapts to many downstream tasks

Text: GPT-4, Claude, Gemini — trained on internet text, generalize to any language task

Vision: CLIP, DALL·E — trained on image-text pairs, generalize to visual understanding

Key idea: learn general capabilities once, apply everywhere without retraining

The question: can this paradigm work for tabular data?

What Are Tabular Foundation Models?



Traditional ML learns parameters for YOUR data. A TFM has learned HOW to learn from ANY tabular data.

Main Paper: TabPFN-2.5

TabPFN-2.5: Advancing the State of the Art in Tabular Foundation Models

Yu, Jablonski, Hoo, Garg, Robertson, Bühler et al. — Prior Labs & Univ. of Freiburg, 2025

TabPFN v1 (2022)

1K samples, 100 features
Proof of concept

TabPFNV2 (2024)

10K samples, 500 features
Competitive with XGBoost

TabPFN-2.5 (2025)

50K samples, 2K features
Beats tuned XGBoost

Architecture & Method

Alternating Attention: row-wise (inter-sample) $\leftrightarrow$ column-wise (inter-feature) layers

24 transformer layers for classification, 18 for regression

Feature Group Embeddings: groups of features embedded together $\rightarrow$ faster inference

64 'Thinking' Rows: learned dummy inputs as computational scratch space (inspired by CoT in LLMs)

Pre-trained on synthetic data; Real-TabPFN-2.5 fine-tuned on 43 curated real datasets

Distillation engines: convert to fast MLP or Tree Ensemble for production deployment

Benchmarks & Metrics

TabArena

51 diverse real-world datasets
Up to 100K rows, 2K features

Internal Suite

100+ proprietary datasets
Healthcare, finance, manufacturing

RealCause

Causal inference benchmark
Treatment effect estimation

Metrics

Classification: Accuracy, ROC AUC, Normalized Score (0–1 per dataset)

Regression: RMSE, R^2

Causal Inference: PEHE (Precision in Estimating Heterogeneous Effects)

Key Results from the Paper

TabPFN-2.5 (default) > tuned XGBoost, CatBoost, LightGBM

Single forward pass, zero hyperparameter tuning

TabPFN-2.5 $\approx$ AutoGluon 1.4 (4-hour extreme mode)

Matches a massive multi-model ensemble run for 4 hours

Real-TabPFN-2.5 > everything

Fine-tuned variant sets new SOTA on classification

Causal inference SOTA on RealCause benchmark

Best CATE estimation as T-Learner base model

Ablation Studies

▶ Deeper network (24 layers)

→ Better accuracy on complex datasets

▶ Larger feature groups

→ Faster training & inference

▶ 64 'Thinking' rows

→ Extra compute capacity — measurable gains

▶ Richer synthetic priors

→ Better generalization to diverse data

▶ Scale to 50K rows

→ 5× more real-world problems covered

▶ Real-data fine-tuning

→ Significant accuracy boost over synthetic-only

My Reproduction Setup

Datasets

Breast Cancer Wisconsin — 569 samples, 30 features, 2 classes

Wine — 178 samples, 13 features, 3 classes

Models

Logistic Regression (linear baseline)

Random Forest (200 trees)

XGBoost (200 estimators, lr=0.1)

TabPFN (16 estimators, GPU)

Protocol

5-fold stratified cross-validation • Metrics: Accuracy & Weighted F1 • No hyperparameter tuning for any model

My Reproduction Results

Breast Cancer

Logistic Reg.	0.974
Random Forest	0.954
XGBoost	0.963
TabPFN	0.977 ★

Wine

Logistic Reg.	0.983
Random Forest	0.978
XGBoost	0.961
TabPFN	0.983 ★

TabPFN achieves the highest accuracy on Breast Cancer and ties for the top on Wine — with zero tuning. Confirms paper's claim: tabular foundation models are strong defaults on small-to-medium datasets.

What I Learned

TabPFN-2.5 is not a toy — it's a practical tool ready to replace XGBoost as a default baseline

A single forward pass beating a 4-hour AutoGluon ensemble is genuinely remarkable

'Thinking rows' = chain-of-thought for tabular data — elegant cross-pollination from LLMs

The ecosystem (distillation, API, interpretability) matters as much as the model itself

We are witnessing convergence: text, vision, and now tabular all moving to foundation models

Limitations

Dataset size ceiling: 50K rows max — many production datasets have millions

GPU memory: 24-layer transformer on 50K rows needs serious hardware

Proprietary distillation: fast deployment engines are commercial, not open-source

Interpretability gap: tree models offer SHAP/feature importance; transformers are harder to inspect

Not universal: on very large, well-tuned datasets, XGBoost ensembles may still compete

Conclusion

For Practitioners

If your dataset has < 50K rows, TabPFN should be your first model to try

For Researchers

Can we scale to millions of rows? Combine tabular + text in one model?

The Big Picture

The age of tabular foundation models is here — XGBoost finally has serious competition

References

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github.com/ArshanBhanage/Deep-Learning---Short-Story